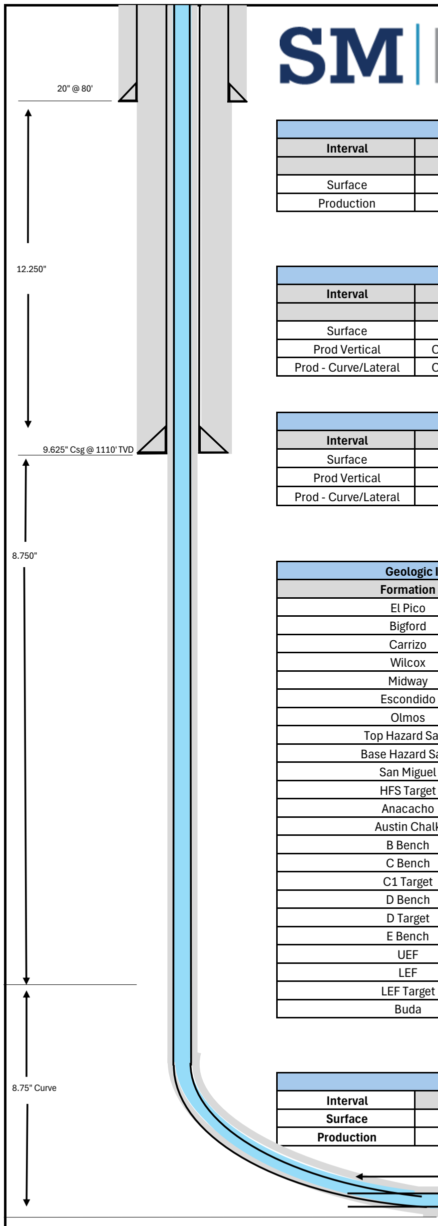

SM | ENERGY

San Ambrosia A (SA5) 4018H

Rig: Nabors X19	Webb County	Austin Chalk - C
SHL LAT: 28.179078 / LONG: -100.093243		
API: 42-479-45551	Permit: 912994	AFE: 3205224
KB+GL/GL: 616.6' / 584.1'	Target: 5,580' TVD @ -103' VS w/ 90° Inc	

Casing Design

Interval	Depth	Hole Size	Csg Size	Weight	Grade	Conn.	ID	Drift	Collapse	Burst	Coupling OD
	ft	in	in	#/ft			in	in	psi	psi	in
Surface	1,110'	12.25"	9.625	36	J-55	BTC	8.921	8.765	1616	2816	9.625
Production	18,341'	8.75"	5.5	23	P110RY	USS TALON - HTQ	4.67	4.545	12248	11616	5.95

Drilling Fluids Information						Provider: Baroid					
Interval	Fluid	Depth	Weight	BHP (min)	Visc	PV	YP	PH/OWR	WPS	Scavenger	% LGS
		ft	#/gal	#/gal	sec/qt	cP	#/100ft^2		ppm	/Inhibitor	
Surface	WBM	1,110'	8.5 - 9.0	N/A	28-44	2 - 7	2 - 12	7 - 9	275k - 300k	NA	<6%
Prod Vertical	OBM - INVERMUL	1,110' - 5,194'	11 - 11.9	N/A	40 - 44	2 - 4	8 - 14	78/22 - 85/15	275k - 300k	0	<10%
Prod - Curve/Lateral	OBM - INVERMUL	5,194' - 18,341'	11 - 11.9	N/A	40 - 44	2 - 4	8 - 14	78/22 - 85/15	275k - 300k	0	<10%

Bit Details						Motor Details				
Interval	Size	Mfg.	Model	TFA	Material	Size	Bend	Lobe/Stage	RPG	Notes
Surface	12.25	Halliburton	GTF650	1	Steel	8-3/4" x 10" 7/8 7.0	1.83	7/8	0.18	11 3/4" NBS & 11 3/4" String Stab
Prod Vertical	8.75	Reed	TK66 CO1	1	Matrix	7" 7/8 6.9	1.83	7/8	0.25	8 1/2" NBS & 8 1/4" String Stab
Prod - Curve/Lateral	8.75	Reed	TK63 G9	1.5	Steel	7" 5/6 9.5	2	5/6	0.34	8 1/2" NBS/String Stab - Agitator 3000' Back

Cement Details		Provider: Halliburton			
Interval	Slurry	Weight	TOC/VOL	Excess	Notes
Surface	Spacer	8.3	40 Bbls	N/A	Nine Cement
	Lead	12.8	0'	80%	
	Tail	14.8	888'	20%	
Production	Spacer	13	100 Bbls	N/A	Halliburton Cement
	Lead 1	13.5	0'	5%	
	Lead 2	15.5	1376.6'	5%	
	Tail	15.5	5094'	5%	

Geologic Information

Formation	TVD
El Pico	surface
Bigford	surface
Carrizo	surface
Wilcox	surface
Midway	929
Escondido	1,166
Olmos	2,377
Top Hazard Sand	2,733
Base Hazard Sand	3,329
San Miguel	3,638
HFS Target	3,877
Anacacho	4,809
Austin Chalk	5,060
B Bench	5,210
C Bench	5,307
C1 Target	5,395
D Bench	5,519
D Target	5,599
E Bench	5,731
UEF	5,816
LEF	6,000
LEF Target	6,107
Buda	6,175

Directional Overview

MD Build Start	1,110'
MD Build End	5,194'
Max Build Rate °/100'	2°/100'
Max Tangent °	15°
Stepout [ft]	892'
KOP MD	5,194'
KOP TVD	5,087'
EOC MD	5,976'
EOC TVD	5,580'
Curve Build °/100'	12°/100'
EOL MD	18,341'
EOL TVD	5,530'
Lateral Length	12,365'
Lateral Inc	90.14°
Lateral Azimuth	239.24°
BHL ' EW from LL	N/A
BHL ' NS from LL	130' FNLL

Temperature Data

Csg String	Depth Reference [ft TVD]	BHST [°F]
Surface	Top of Tail Slurry	96
	Surface CSG Shoe	100
Production	Top of Lead #2 Slurry	105
	Top of Tail Slurry	173
	Production Csg Shoe	181

Casing Jewelry

Interval	Centralizers
Surface	Red-X 9 5/8" x 12 1/4" Bow Spring Centralizers: 1 every joint in shoe track, 1 every other joint F/ Top of Shoe Track T/Surface
Production	Swell packer 500' Inside Sfc Csg. Arsenal Floatation Sub 100' after LP. 8.25" Red-X Bow Spring every jt F/ TD T/KOP. 8.25" Legend SSB every jt F/KOP T/100' Inside Sfc Csg. Legend SSB Every other jt T/surface